

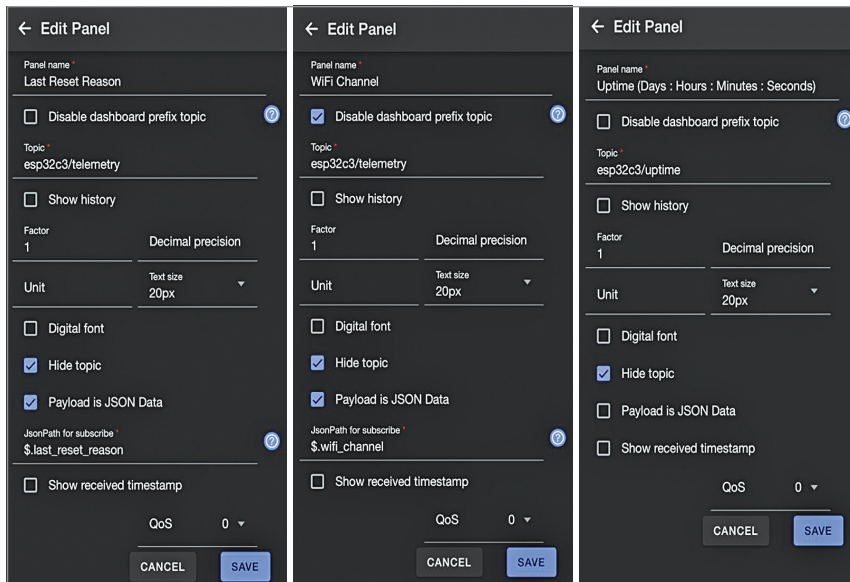


Fig. 16: Configuration of the last reset reason, Wi-Fi channel, and uptime telemetry panels in the IoT MQTT Panel app



Fig. 18: IoT MQTT Panel dashboard displaying real-time ESP32-C3 Super Mini telemetry, onboard LED control, LED status, and remote reset functionality

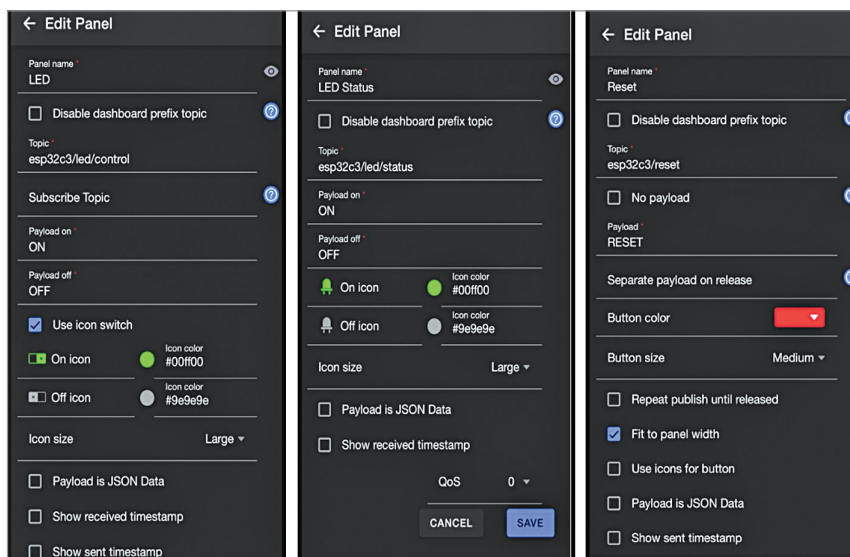


Fig. 17: Configuration of the LED control, LED status, and reset control panels in the IoT MQTT panel app

system stability and reliability while detecting unexpected resets or power interruptions. Fig. 10 shows the function used to calculate and format the ESP32-C3 uptime in DDD:HH:MM:SS format.

Remote LED status

The device publishes the LED status to the MQTT broker, ensuring that the dashboard always reflects the actual hardware state, even after a restart or a remote command. Fig. 11 shows the function used to

publish the LED on/off status to the MQTT broker.

Remote LED control (on/off)

The dashboard enables remote control of the ESP32-C3 Super Mini's onboard LED through MQTT messages. The ESP32-C3 subscribes to the LED control topic and updates the LED state whenever an on or off command is received. After changing the LED state, the device publishes the updated status to the dashboard, ensuring that

the displayed state always matches the actual hardware state. Fig. 12 shows the MQTT callback function used to receive LED control commands and operate the onboard LED.

Note. The ESP32-C3 Super Mini's onboard LED is connected to GPIO 8 and uses active-low logic. Setting GPIO 8 low turns the LED on, while setting GPIO 8 high turns it off.

Remote ESP reset

The dashboard enables the ESP32-C3 Super Mini to be restarted remotely by sending an MQTT reset command. When the device receives RESET message on the subscribed reset topic, the firmware executes the `ESP.restart()` function, causing the microcontroller to reboot. Fig. 13 shows the callback function used to receive the MQTT reset command and restart the ESP32-C3 Super Mini. This feature is useful for remotely recovering devices that become unresponsive without requiring physical access.